

Fault Detection, Isolation, and Diagnosis

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Introduction and Motivation

- Modern engineering systems—aerospace systems, chemical processes, power networks, manufacturing lines—are **safety-critical**.
- There is an ever-increasing demand for **reliability** and **safety** in industrial systems subject to potential process abnormalities and component faults.
- It is important to:
 - ① **Detect and identify** abnormalities and faults as early as possible.
 - ② Implement **fault-tolerant operation** to **minimize performance degradation and avoid dangerous situations**.
- An undetected fault in a production line can eventually require an **overall plant shutdown**, which **can be extremely costly**.

Central Challenge

Environmental noise, inevitable modelling errors, and the complexity of system dynamics make analytical fault diagnosis significantly more challenging than hardware-redundancy approaches.

What Is a Fault?

Formal Definition

A **fault** is defined as an *unpermitted deviation* of *at least one* characteristic property or parameter of the system from the acceptable, usual, or standard condition.

Examples of faults include:

- Blocking of an actuator.
- *Loss of a sensor* (e.g., a sensor stuck at a particular value or a variation in the sensor scalar factor).
- *Disconnection of a system component*.

Classification of faults:

- **Actuator faults:** Interrupt the control action from the controller to the plant.
- **Sensor faults:** Produce substantial measurement errors.
- **Plant (component/parameter) faults:** Directly change the dynamic input/output properties of the system.

Fault Diagnosis: Three Core Tasks

1. Fault Detection

The most basic task: determine *whether* a malfunction or fault exists in the system and *when* it occurs.

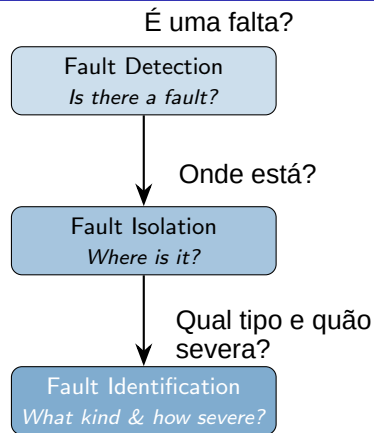
2. Fault Isolation

Determine the *location* of the faulty component within the system.

3. Fault Identification (Diagnosis)

Determine the *type*, *shape*, and *size* (severity) of the fault.

Note: The term “fault diagnosis” is often used as a synonym for the combined FDI process. Some authors also use **FDIR** (Fault Detection, Isolation, and Recovery).



Prognostics and Health Management (PHM)

Definition of Prognostics

Prognostics is the process of predicting the future health state of a system or component, typically expressed as the *Remaining Useful Life (RUL)*—i.e., the time left before a system can no longer perform its intended function.

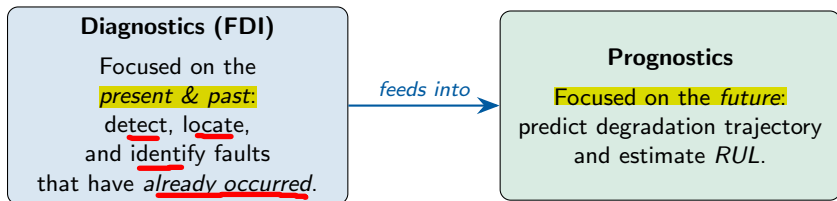
Prognostics and Health Management (PHM)

PHM provides methodologies and tools for creating **tailored maintenance plans** based on the specific characteristics, operations, and degradation scenarios of a given asset. Its objective is to achieve **optimal system availability while minimizing costs**.

PHM integrates:

- 1 **Fault Detection and Isolation (FDI)** — current state awareness.
- 2 **Health Management** — decision-making and maintenance planning.
- 3 **Prognostic Capabilities** — RUL prediction for forward-looking insight.

FDI vs. Prognostics: A Conceptual Contrast



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Diagnostics asks:

- Is there a fault?
- Where is the fault?
- What type and severity?

Prognostics asks:

- How is the system degrading?
- How much useful life remains?
- When should maintenance occur?

Hardware vs. Analytical Redundancy

Hardware Redundancy

- Uses **identical physical components** with the same input signal.
- Duplicated output signals are compared via limit checking or majority voting.
- **Pro:** Highly **reliable**.
- **Con:** **Expensive; increases weight and occupies more space.**

Analytical (Software) Redundancy

- Uses **mathematical models or data-driven algorithms** instead of **duplicate hardware**.
- **Checks consistency of real-time process data against prior knowledge of healthy system behaviour.**
- **Pro:** Cost-effective; no additional physical hardware.
- **Con:** Sensitive to modelling errors, noise, and system complexity.

Historical Note

Analytical redundancy for FDI was originated by R. V. Beard in his 1971 MIT dissertation, introducing observer-based methods for failure accommodation in linear systems.

Model-Based Fault Diagnosis — Overview

Core Principle

A **mathematical model** of the system (obtained via physical principles or system identification) is used to monitor the *consistency* between measured outputs and model-predicted outputs. Discrepancies generate **residual signals** used for fault detection, isolation, and identification.

General residual generation scheme:

$$r(k) = y(k) - \hat{y}(k)$$

where $y(k)$ is the measured output and $\hat{y}(k)$ is the model-predicted output.

Key design objective: Make $r(k)$ *sensitive* to faults but *robust* against disturbances and modelling errors:

$$\min_K \left(\frac{\|G_{\bar{d}}(z)\|}{\|G_{\bar{f}}(z)\|} \right)$$

where $G_{\bar{d}}$ is the transfer from disturbances to residual and $G_{\bar{f}}$ is the transfer from faults to residual.

Model-Based FDI — Major Techniques

1 Observer-Based Methods

- Luenberger observers, proportional–integral (PI) observers, sliding-mode observers, descriptor observers, unknown input observers.
- Banks of observers enable **fault isolation** via structured or generalized residuals.

2 Parity Relation Approach

- Generates a *parity vector* to check consistency between model and process outputs.
- Equivalent to the use of a dead-beat observer.

3 Parameter Estimation Methods

- Online identification of system parameters (e.g., least squares), compared with reference parameters from healthy conditions.
- Faults assumed to be reflected in parameter changes.

4 Kalman-Filter-Based Methods (stochastic systems)

- EKF, UKF, adaptive/augmented-state Kalman filters.
- Statistical testing of residuals (innovations): χ^2 test, generalized likelihood ratio, CUSUM.

Model-Based FDI — Advantages and Limitations

Advantages

- Strong theoretical foundation in control theory.
- Well-defined residual generation and evaluation framework.
- Capable of detecting, isolating, and identifying faults simultaneously.
- Applicable to a wide range of systems (linear, nonlinear, stochastic, hybrid, networked).

Limitations

- Requires an **accurate system model**, which may be difficult or expensive to obtain.
- Performance degrades with **modelling errors** (ΔA , ΔB , ΔC).
- Complexity increases for large-scale, highly nonlinear, or time-varying systems.
- Sensitive to unknown disturbances and noise.

Model-Free FDI — Signal-Based Methods

Core Principle

Signal-based methods utilize **measured signals directly**— rather than explicit input–output models—for fault diagnosis. Features are extracted from sensor data, and diagnostic decisions are made via *symptom/pattern analysis* against prior knowledge of the healthy system.

Three sub-categories:

- **Time-Domain:** Mean, variance, RMS, peak values, trends, slope, phase angles.
Example: Monitoring RMS current to detect transistor faults in power converters.
- **Frequency-Domain:** Spectral analysis via DFT/FFT.
Example: Motor Current Signature Analysis (MCSA) for broken rotor bars.
- **Time–Frequency Domain:** STFT, Wavelet Transform, Hilbert–Huang Transform, Wigner–Ville Distribution.
Example: Tracking fault frequency components under non-stationary operating conditions.

Model-Free FDI — Knowledge-Based (Data-Driven) Methods

Core Principle

When only a **large volume of historical data** is available (rather than a model or signal pattern), artificial intelligence techniques are used to extract underlying knowledge that implicitly represents the dependence among system variables.

Qualitative methods:

- **Expert systems** — rule-based reasoning from human expertise.
- **Qualitative Trend Analysis (QTA)** — identification of process trends from noisy data.

Quantitative methods:

- **Statistical:** PCA, PLS, ICA, SVM.
- **Non-statistical:** Neural Networks (various topologies), Fuzzy Logic.
- **Joint approaches:** e.g., Bayesian Network + NN, dynamic PCA + feedforward NN.

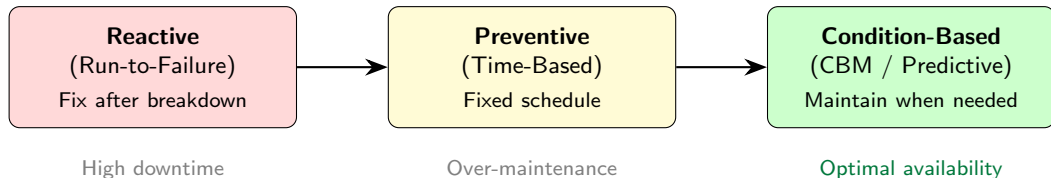
Model-Based vs. Model-Free FDI

Criterion	Model-Based	Model-Free (Signal / Knowledge)
Requirement	Accurate mathematical model	Measured signals or large historical dataset
Residual generation	Explicit: $r = y - \hat{y}$ via observers, parity, etc.	Implicit: feature extraction & pattern matching
Fault isolation	Structured/generalised residual banks	Classification (SVM, NN, expert rules)
Robustness to model error	Sensitive—requires robust design	Inherently model-free; depends on data quality
Interpretability	High (physics-based insight)	Variable (expert systems: high; deep NN: low)
Scalability	Difficult for very large-scale systems	Scales with data; needs representative training
Historical data needed	Minimal (only model parameters)	Extensive (training / reference datasets)

Hybrid Approaches

In practice, **hybrid methods** combine two or more approaches (e.g., model-based residuals classified by neural networks) to leverage complementary strengths.

From Reactive to Condition-Based Maintenance



Condition-Based Maintenance (CBM)

CBM relies on **real-time monitoring data** and **diagnostic/prognostic analysis** to schedule maintenance actions only when evidence of degradation or incipient faults is observed—achieving optimal system availability while minimizing costs.

Automating CBM: The PHM Framework

Step 1: Data Acquisition — Sensors, SCADA, IoT devices collect operational data



Step 2: Feature Extraction — Time, frequency, or time–frequency domain signal processing



Step 3: Fault Detection — Identify that something is going wrong; determine fault onset time



Step 4: Fault Isolation & Identification — Locate and characterize the fault (type, severity)



Step 5: Prognostics — Predict degradation trajectory and Remaining Useful Life (RUL)



Step 6: Decision Support — Automated maintenance scheduling, fault-tolerant reconfiguration

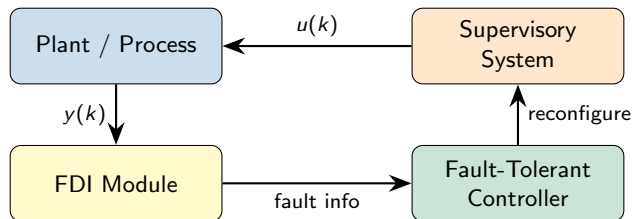
Fault-Tolerant Control: Closing the Loop

Principle

Fault-tolerant control (FTC) is **integrated with fault diagnosis** in essence. Real-time FDI provides the information needed for the supervisory system to take appropriate corrective actions automatically.

Fault-tolerant responses include:

- **Signal compensation:** Offset faulty actuator/sensor signals.
- **Controller tuning or reconfiguration:** Adapt controller parameters or switch control law.
- **Component replacement:** Activate redundant hardware duplicates when available.



Hybrid and Active Fault Diagnosis

Hybrid Fault Diagnosis

An **integration or combination** of more than one diagnosis method to leverage complementary strengths:

- Model-based residuals + neural-network classifiers.
- Signal features + SVM or fuzzy-logic decision.
- PCA feature extraction + expert-system reasoning.

Hybrid methods often achieve **better overall performance** than any single method alone.

Active Fault Diagnosis

Enhances detectability by **injecting a suitably designed test input signal** so that:

- Faulty modes can be distinguished from normal modes *quickly and accurately*.
- The system response to the excitation provides additional diagnostic information.
- Particularly useful when faults are otherwise difficult to detect under normal operating conditions.

Taxonomy Summary

Fault diagnosis methods: **Model-based** | **Signal-based** | **Knowledge-based** | **Hybrid** | **Active**.

Summary of Key Concepts

- ➊ **Fault Diagnosis (FDI)** encompasses three sequential tasks—detection, isolation, and identification—that together provide situational awareness of system health.
- ➋ **Prognostics** extends FDI into the future by predicting degradation trajectories and Remaining Useful Life, enabling proactive maintenance decisions.
- ➌ **Model-based FDI** relies on mathematical system models and residual analysis; it offers strong theoretical guarantees but requires accurate models.
- ➍ **Model-free FDI** (signal-based and knowledge-based) operates directly on measured or historical data; it avoids modelling requirements but needs representative datasets.
- ➎ **Condition-Based Maintenance (CBM)**, enabled by the PHM framework, automates maintenance scheduling by integrating FDI and prognostics—achieving optimal availability at minimum cost.
- ➏ **Fault-tolerant control** closes the loop by using diagnostic information to reconfigure the system in real time.
- ➐ **Hybrid and active approaches** combine multiple methods or use test excitations to improve diagnostic performance.

Does it make sense?

Questions & Discussion